

Azure Networking Essentials Training Manual

This manual covers the core concepts of Azure networking including Virtual Networks (VNETs), Subnets, CIDR ranges, VNet Peering, connectivity alternatives, and Azure Load Balancer types. It is designed for cloud professionals and students preparing for Azure certifications or practical deployments.

1. Azure Virtual Network (VNet)

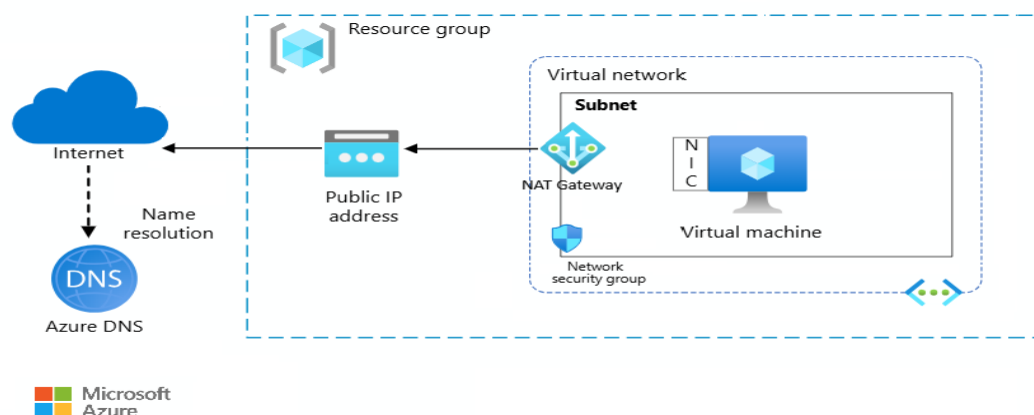
What is a VNet?

- A logically isolated network in Azure, similar to a traditional on-premises network.
- Hosts Azure resources such as Virtual Machines, databases, and services.
- Provides secure communication within Azure and with on-premises networks.

Key Features

- Customizable IP address ranges.
- Segmentation using subnets.
- Integration with on-premises networks via VPN or ExpressRoute.

Diagram



2. Subnets & CIDR in Azure

Subnets

- Divide a VNet into smaller address spaces for resource grouping and isolation.

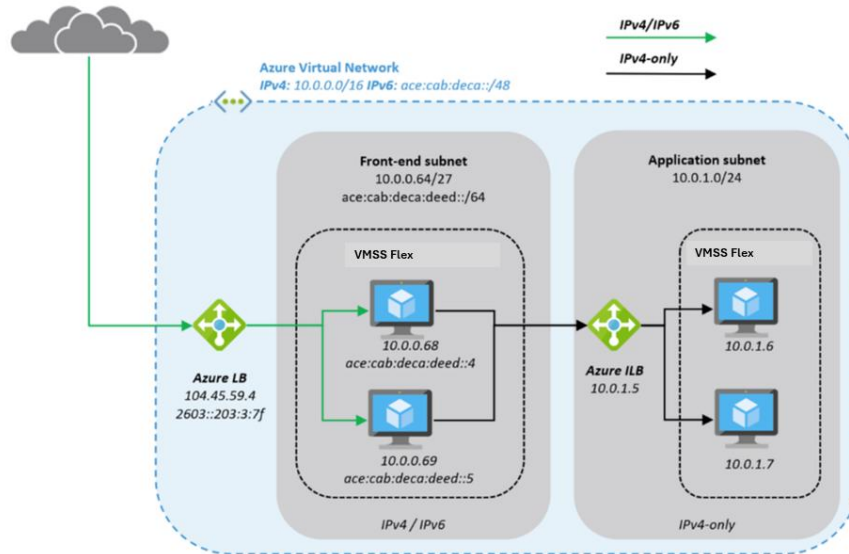
CIDR (Classless Inter-Domain Routing)

- Allowed range: /2 to /29.
- Example: VNet 10.0.0.0/16 can have subnets like 10.0.1.0/24, 10.0.2.0/24.

Rules

- Subnet ranges must be within the VNet's CIDR.
- No overlap between subnets.
- Each subnet can host Network Security Groups (NSGs), route tables, and services.

Diagram



3. VNet Peering

What is VNet Peering?

- Connects two VNets so they act as one for communication.

Types

- Local Peering: Within the same region.
- Global Peering: Across different regions.

History

- Introduced in 2016, with later expansion to Global Peering.

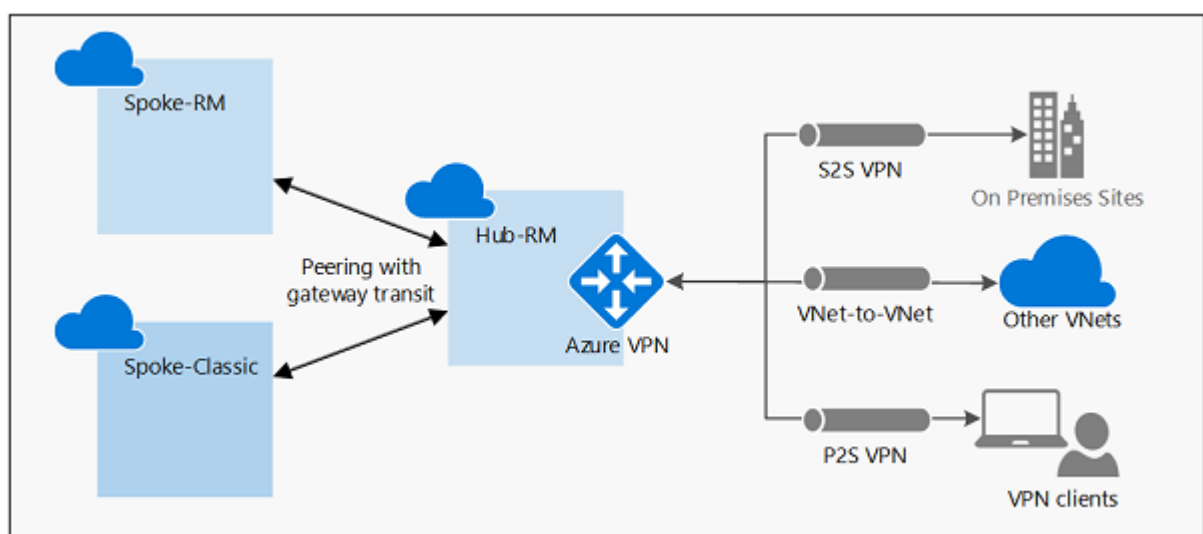
Do's

- Use peering for low-latency, high-bandwidth connections.
- Ensure non-overlapping CIDR ranges.
- Apply NSGs and route tables for traffic control.
- Use hub-and-spoke architecture for scalability.

Don'ts

- Don't rely on peering for transitive routing.
- Don't overlap IP ranges.
- Don't expect encryption by default.
- Avoid peering across subscriptions without permissions.

Diagram



Azure VNet Peering hub and spoke model diagram

4. Alternatives to VNet Peering

- VPN Gateway: Secure IPsec/IKE tunnel between Azure and on-premises.
 - ExpressRoute: Private, dedicated connection bypassing the public internet.
 - Azure Virtual WAN: Centralized hub for connecting multiple sites, VNets, and remote users.
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5. Azure Load Balancers

Azure offers multiple load balancing solutions depending on use case:

Type	Scope	Features	Use Case
Basic Load Balancer	Single availability set	Free, limited features	Small dev/test workloads
Standard Load Balancer	Across availability zones	High availability, secure, supports HA ports	Production workloads
Application Gateway (Layer 7)	HTTP/HTTPS traffic	SSL termination, WAF, URL-based routing	Web applications
Traffic Manager (DNS-based)	Global	Distributes traffic across regions	Geo-distribution, failover
Front Door	Global edge network	CDN + WAF + SSL offload	High-performance global web apps

Diagram

 Application Gateway	 Load Balancers	 Traffic Manager	 Front Doors
Works at Layer 7 OSI model	Works at Layer 4 OSI model	DNS Based traffic load balancer	Offers Layer 7 capabilities
Regional load balancing	Global load balancing	Global load balancing	Global load balancing
HTTP, HTTPS	TCP & UDP protocols	DNS resolutions	HTTP, HTTPS and HTTP/2
Sticky session supported	Sticky session supported	Sticky session not supported	Sticky session supported

Azure Load Balancer types comparison chart

6. Key Takeaways

- CIDR range: /2 to /29 allowed.
- Peering: Best for intra-cloud, low-latency connections; introduced in 2016.
- Alternatives: VPN Gateway, ExpressRoute, Virtual WAN.
- Load Balancers: Choose based on scope—Basic/Standard for infra, Application Gateway for Layer 7, Traffic Manager/Front Door for global distribution.

References

1. Azure Virtual Network Guide: VNets, Subnets and NSGs
2. Azure Virtual Network Peering | Azure Docs
3. Configure subnet peering - Azure Virtual Network

Azure Networking Essentials Guide
